

ENOCK KUMI ACKAAH

Applied Statistics | Machine Learning | Biomedical Data Science

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EDUCATION

Georgia State University, Atlanta, Georgia <i>M.S. in Applied Statistics, Concentration in Machine Learning and Data Science</i>	Expected May 2028
The University of Akron, Akron, Ohio <i>M.S. in Statistics, GPA: 3.5/4.00</i> – Research advisor: Professor Sujay Datta	May 2026
University of Cape Coast, Cape Coast, Ghana <i>B.Sc. in Mathematics and Statistics, GPA: 3.16/4.00</i>	August 2023

PUBLICATIONS AND RESEARCH SOFTWARE

- Ackaah, E. K., & Datta, S. (2026). *Cancer Classification in Low- and High-Dimensional Biomedical Data Using Machine Learning Models*. Preprints. <https://doi.org/10.20944/preprints202608.0073.v1>
- Ackaah, E. K., & Datta, S. (2026). *Artificial Intelligence in the Biomedical Sciences and Healthcare—Its Impact, Potential Benefits and Pitfalls*. Manuscript in preparation for submission to *Precision Oncology*.
- Ackaah, E. K. (2026). *Statistical Reliability of High-Dimensional ER-Status Prediction Models* (Version 0.1.1) [Computer software]. Zenodo. doi:10.5281/zenodo.21524693.
- Ackaah, E. K. (2026). *Frozen 72-CpG DNA Methylation Classifier for Colorectal Tumor Detection* (Version 1.0.0) [Computer software]. Zenodo. doi:10.5281/zenodo.21819138.

RESEARCH EXPERIENCE

Artificial Intelligence in Biomedical Sciences and Healthcare <i>Narrative review with structured thematic literature search</i> Co-author and research supervisor: Professor Sujay Datta	June 2026–Present
– Synthesized evidence across machine learning, deep learning, multimodal AI, drug discovery, clinical decision support, and human–AI collaboration. – Evaluated reliability risks including small samples, data leakage, weak external validation, limited explainability, bias, privacy, governance, and implementation barriers. – Developed a framework emphasizing reproducibility, transparent reporting, clinically meaningful evaluation, human oversight, and prospective monitoring.	
Statistical Reliability of High-Dimensional Breast Cancer RNA Sequencing Models <i>Independent research project; manuscript draft and archived software release</i> Development cohort: GSE81538; external-validation cohort: GSE96058 GitHub repository Zenodo v0.1.1: doi:10.5281/zenodo.21524693	July 2026
– Built a leakage-safe Python/scikit-learn pipeline for 397 development, 405 sensitivity, and 3,073 external-validation tumors across 18,213 aligned genes. – Assessed prediction and feature stability across repeated splits and sample sizes; mean Jaccard similarity reached 0.7693, with near-random failure at $n = 50$. – Achieved external AUROC 0.9699 (95% CI: 0.9597–0.9788) and balanced accuracy 0.9390; the sensitivity model achieved AUROC 0.9672 and Brier score 0.0510.	
Cancer Classification in Low- and High-Dimensional Biomedical Data <i>The University of Akron</i> Research advisor: Professor Sujay Datta	September 2025–July 2026
– Compared logistic regression, SVM, random forest, and KNN on a 106-sample breast-tissue dataset and a 15-sample osteosarcoma DNA-methylation dataset with approximately 480,000 CpG features. – Applied PCA to the methylation data; eight components explained 81.6% of variance, with sensitivity analyses across train–test splits and component choices. – Observed stable low-dimensional performance (random forest and linear SVM accuracy 0.7895) but substantial high-dimensional small-sample instability; led analysis and manuscript drafting.	

SELECTED GRADUATE PROJECTS

Selected Graduate Research Projects <i>The University of Akron</i>	2024–2026
– Fit and assessed Cox proportional-hazards models for the GBSG2 breast-cancer dataset. – Used exploratory, permutation-based, and multivariate methods to analyze neuropsychological brain-activity data in R. – Analyzed clinical biomarkers in SAS/SPSS and modeled educators’ confidence in AI integration.	

TEACHING AND ACADEMIC APPOINTMENTS

Graduate Teaching Assistant, Department of Statistics

August 2024–May 2026

The University of Akron

Akron, Ohio

- Led weekly statistics laboratories for approximately 15–20 students, integrating statistical concepts with software-based analysis.
- Tutored students in statistics, probability, R, SPSS, Minitab, and quantitative problem solving.
- Graded assignments and assessments for large sections and provided detailed feedback to support learning.

Teaching Assistant

September 2023–July 2024

University of Cape Coast

Cape Coast, Ghana

- Tutored undergraduate students in statistics and quantitative methods and supported grading and instructional activities.

STATISTICAL CONSULTING AND PROFESSIONAL EXPERIENCE

Graduate Statistical Consulting

Spring 2026

The University of Akron

Faculty supervisor: Dr. Jun Ye

- Translated an applied research question into an analytical plan, interpreted results, and communicated findings in a team presentation.

SELECTED PRESENTATIONS

- Ackaah, E. K. *Cancer Classification in Low- and High-Dimensional Data: A Comparative Study Using Supervised and Semi-Supervised Machine Learning*. Master's research defense, The University of Akron, May 2026.
- Ackaah, E. K., et al. *Statistical Consulting Project Presentation*. Graduate Statistical Consulting, The University of Akron, Spring 2026.
- Ackaah, E. K., et al. *Lung Cancer*. Graduate coursework presentation, The University of Akron, Fall 2025. doi:10.13140/RG.2.2.31418.25282.

TECHNICAL SKILLS

- Programming and computing: Python, R, SQL, C++, Git, GitHub, Jupyter Notebook, and VS Code.
- Machine learning: logistic and multinomial regression, SVM, random forest, KNN, Lasso/Elastic Net, feature selection, and PCA.
- Validation and reliability: repeated cross-validation, bootstrap confidence intervals, external validation, calibration, Brier score, and feature-stability analysis.
- Statistical methods: linear models, survival and Cox analysis, Bayesian methods, multivariate and nonparametric analysis, experimental design, and permutation tests.
- Biomedical data and software: RNA sequencing, DNA methylation, high-dimensional omics, GEO, scikit-learn, pandas, NumPy, SciPy, SAS, SPSS, Minitab, Excel, Power BI, and Tableau.

PROFESSIONAL MEMBERSHIPS

- American Statistical Association